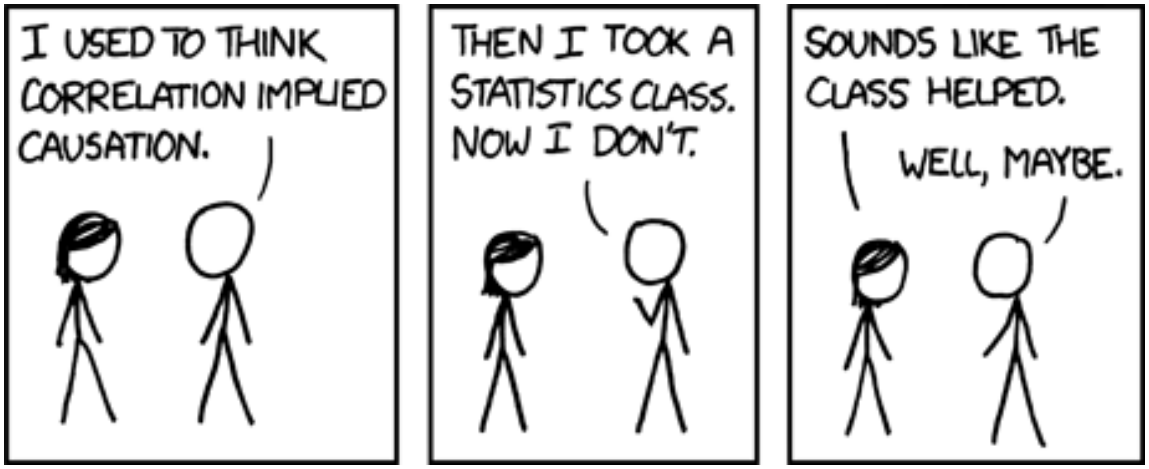
Advanced Applied Econometrics

Teacher: Felix Weinhardt

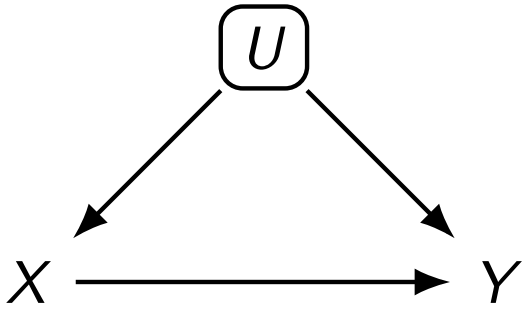


- Last week: Frish-Waugh/Regression Anatomy, OVB, AET2005 and Oster-method
- Today:
- Heterogeneous effects in OLS
- Fisher Inference

Standard Interpretation of Coefficient Changes

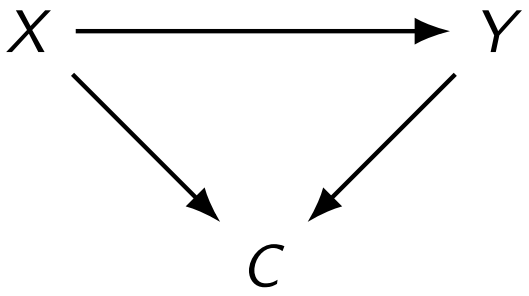
- Add controls $\rightarrow$ coefficient changes
- Standard interpretation:
  - Omitted variable bias
- Based on causal DAG reasoning

Confounder Bias (Omitted Variable)



- U affects both X and Y
- Omitting $U \rightarrow$ biased estimate
- **Control for U (boxed variable)**

Collider Bias (Bad Control)



- C is a collider
- Conditioning on C induces correlation
- Creates bias

Takeaway from Standard DAGs

Usual Logic

Coefficient movements reflect:

- Removing bias (confounders)
- Introducing bias (colliders)

Implicit Assumption

Single treatment effect

New Point

- Even without confounders or colliders:
- Coefficients can still change
- **Reason: heterogeneous treatment effects**

Heterogeneity DAG

$$W \longrightarrow X \longrightarrow Y$$

- W affects X, not Y
- No confounding, no collider
- Still: coefficients change

Why?

- Controls change variation in X
- OLS weights depend on that variation
- $\rightarrow$ **reweighting across groups**

OLS with Heterogeneity

$$\beta^{OLS} = \sum_g w_g \beta_g$$

- $w_g \propto \pi_g \cdot Var(X|g)$

Adding Controls

$$\beta = \frac{Cov(\tilde{X}, Y)}{Var(\tilde{X})}$$

- $\tilde{X}$ depends on controls
- $\Rightarrow$ weights change

Numerical Example 1: Data Generating Process

- Two groups $g \in \{1, 2\}$, each with 50% probability
- Outcome equation:

$$Y_i = \beta_g X_i + \varepsilon_i$$

- Heterogeneous effects:

$$\beta_1 = 1, \quad \beta_2 = 5$$

- Key feature:
  - $Var(X|g = 2) > Var(X|g = 1)$

Example 1: What Do We Estimate?

- Separate regressions:

$$Y_i = \beta_g X_i + \varepsilon_i \quad \text{within each group}$$

- Pooled regression:

$$Y_i = \beta X_i + \varepsilon_i$$

- Also compute:
  - Simple average effect:

$$\beta^{AE} = 0.5 \cdot \beta_1 + 0.5 \cdot \beta_2 = 3$$

Example 1: Results

	(1) $\hat{\beta}_1$	(2) $\hat{\beta}_2$	(3) $\hat{\beta}^{AE}$	(4) $\hat{\beta}^{OLS}$
X	0.96	4.98	2.98	4.00

- Columns (1)-(2): recover true group effects
- Column (3): simple average ≈ 3
- Column (4): pooled OLS ≈ 4

Example 1: Interpretation

- Why is OLS so high?
- Because:

$$\beta^{OLS} = \sum_g w_g \beta_g$$

- Weights:

$$w_g \propto Var(X|g)$$

- Group 2 has higher variance $\rightarrow$ gets more weight

Numerical Example 2: Data Generating Process

- Two groups $g \in \{1, 2\}$ with equal probability
- Outcome equation:

$$Y_i = \beta_g X_i + \varepsilon_i$$

- Heterogeneous effects:

$$\beta_1 = 1, \quad \beta_2 = 5$$

- Additional variable:
  - W_i affects X_i
  - W_i has **no direct effect on** Y_i

Example 2: Structure of the Data

- Treatment equation:

$$X_i = \pi_g W_i + \nu_i$$

- Interpretation:
  - W shifts X
  - Strength of relationship can differ across groups

- Outcome:

$$Y_i = \beta_g X_i + \varepsilon_i$$

- Key assumption:

$$E[\varepsilon_i | X_i, W_i] = 0$$

Example 2: Important Implication

- W does not belong in the outcome equation
- Therefore:
  - No omitted variable bias
  - W is an **irrelevant control** for Y
- Standard prediction:
  - Adding W should not change β

Example 2: What Do We Estimate?

- Short regression:

$$Y_i = \beta^S X_i + \varepsilon_i$$

- Long regression:

$$Y_i = \beta^L X_i + \gamma W_i + \varepsilon_i$$

- Using Frisch–Waugh–Lovell:

$$\beta^L = \frac{Cov(\tilde{X}, Y)}{Var(\tilde{X})}$$

Example 2: What Changes?

- Residualized regressor:

$$\tilde{X}_i = X_i - E[X_i|W_i]$$

- This removes variation in X explained by W
- Crucially:
  - This affects groups differently

Example 2: Results

	Without W	With W
OLS	3.00	2.47

- Large change in coefficient
- Despite:
  - No confounding
  - No collider bias

Example 2: Interpretation

- Both estimators are weighted averages:

$$\beta^S = \sum_g w_g^S \beta_g \qquad \beta^L = \sum_g w_g^L \beta_g$$

- Difference:

$$w_g^L \propto Var(\tilde{X}|g)$$
$$w_g^S \propto Var(X|g)$$

- Controls change the weights

Example 2: Final Takeaway

Key Insight

Coefficient change is not bias.
It is a change in the estimand.

- Driven by reweighting across groups
- Even when controls are "irrelevant"

Implications: Augmented OVB Formula

$$\beta^L - \beta^S = \sum_g (w_g^L - w_g^S)\beta_g + \gamma \frac{Cov(W, X)}{Var(X)}$$

Conclusion

- Coefficient movements only bias
- Also reflect:
  - reweighting
  - Each specification speaks to different estimand
  - Implications for model selection and diagnostics
  - We will return to this when discussing the current DiD literature and IV-LATE interpretations

- Fisher inference

Lady tasting tea experiment

- Ronald Aylmer Fisher (1890-1962)
  - Two classic books on statistics: *Statistical Methods for Research Workers* (1925) and *The Design of Experiments* (1935), as well as a famous work in genetics, *The Genetical Theory of Natural Science*
  - Developed many fundamental notions of modern statistics including the theory of randomized experimental design.
- Muriel Bristol (?? - ??)
  - Worked with Fisher at the Rothamsted Experiment Station (which she established) in 1919 (*and a PhD scientist back in the days when women weren't PhD scientists*)
  - During afternoon tea, Muriel claimed she could tell from taste whether the milk was added to the cup before or after the tea
  - Scientists were incredulous, but Fisher was inspired by her strong claim
  - He devised a way to test her claim which she passed. What was the test?

Description of the tea-tasting experiment

- Original claim: Given a cup of tea with milk, Bristol claims she can discriminate the order in which the milk and tea were added to the cup
- Experiment: To test her claim, Fisher prepares 8 cups of tea – 4 **milk then tea** and 4 **tea then milk** – and presents each cup to Bristol for a taste test
- Question: How many cups must Bristol correctly identify to convince us of her unusual ability to identify the order in which the milk was poured?
- Fisher's sharp null: Assume she can't discriminate. Then what's the likelihood that random chance was responsible for her answers?

Choosing subsets

- "8 choose 4" – $\binom{8}{4}$ – ways to choose 4 cups out of 8
  - There are $8 \times 7 \times 6 \times 5 = 1,680$ ways to choose a first cup, a second cup, a third cup, and a fourth cup, in order.
  - There are $4 \times 3 \times 2 \times 1 = 24$ ways to order 4 cups.
- So there are 70 ways to choose 4 cups out of 8, and therefore a 1.4% probability of producing the correct answer by chance

$$\frac{1680}{24} = 70 = 0.014.$$

- Note: the lady performs the experiment by selecting 4 cups, say, the ones she claims to have had the tea poured first.
- For example, the probability that she would correctly identify all 4 cups is $\frac{1}{70}$

Choosing 3

- To get exactly 3 right, and, hence, 1 wrong, she would have to choose 3 from the 4 correct ones.
  1. She can do this by $4 \times 3 \times 2 = 24$ with order.
  2. Since 3 cups can be ordered in $3 \times 2 = 6$ ways, there are 4 ways for her to choose the 3 correctly.
- Since she can now choose the 1 incorrect cup 4 ways, there are a total of $4 \times 4 = 16$ ways for her to choose exactly 3 right and 1 wrong.
- Hence the probability that she chooses exactly 3 correctly is $\frac{16}{70} = \frac{8}{35}$.

Statistical significance

- Suppose the lady correctly identifies all 4 cups.
- Conclusion
  1. Either she has no ability, and has chosen the correct 4 cups purely by chance, or
  2. she has the discriminatory ability she claims.
- Since choosing correctly is highly unlikely in the first case (one chance in 70), we decide for the second.
  1. if she got 3 correct and 1 wrong, this would be evidence for her ability, but not persuasive evidence since the chance of getting 3 or more correct is $\frac{17}{70} = 0.2429$.
  2. by convention, a result is considered statistically significant if the probability of its occurrence by chance is < 0.05, or, less than 1 out of 20.

Null hypothesis

- In this example, the null hypothesis is the hypothesis that the lady has no special ability to discriminate between the cups of tea.
  - We can never prove the null hypothesis, but the data may provide evidence to reject it.
  - In most situations, rejecting the null hypothesis is what we hope to do.
- Randomization allows us to make probability calculations revealing whether the data are "statistically significant" or not.
- Randomization also takes care of all the possible causes for which we cannot control.

Example: Honey experiment

Paul et al (2007) designed a study to evaluate the effect of giving buckwheat honey or honey-flavored destromethorpan or nothing at night before bedtime on nocturnal cough frequency for a population of children with upper respiratory tract infections

- Population: 72 kids (35 received honey, 37 nothing)
- Outcome of interest: "cough frequency afterwards" (*cfa*)
- Pretreatment variable: "cough frequency prior" (*cfp*)

Notation

- Let Y_i^1 and Y_i^0 represent potential outcomes for individual i with and without honey treatment, respectively
- Let $D_i \subset \{0, 1\}$ be a binary indicator equalling 1 if the child received honey as the treatment and 0 otherwise
- Switching equation:

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

- X_i is a covariate/characteristic/pretreatment variable for child i. Here it is cough frequency prior, *cfp*
- Number of treatment (N_t) and control units (N_c):

$$\begin{aligned} N_t &= \Sigma_{i=1}^N D_i \\ N_c &= \Sigma_{i=1}^N (1 - D_i) \end{aligned}$$

Cough frequency for the first six units

Unit	Potential outcomes		Observed variables		
	Y_i^0	Y_i^1	D_i	X_i	Y_i^{obs}
		cfa		*cfp*	*cfa*
1	?	3	1	4	3
2	?	5	1	6	5
3	?	0	1	4	0
4	4	?	0	4	4
5	0	?	0	1	0
6	1	?	0	5	1

Sharp null

- Let $\delta = Y^1 - Y^0$ be the causal effect of the treatment.
- Assess the "sharp null" hypothesis:

$$H_0 : \delta_i = Y_i^1 - Y_i^0 = 0 \text{ for all } i = 1, \ldots, N$$

  against the alternative that for some units there is some non-zero effect of the treatment ($\delta_i \neq 0$)
- **Key feature**: The null hypothesis is considered **sharp** because under the sharp null hypothesis, we know the missing potential outcomes for each observation
- How's that? If $\delta_i = 0$, then we aren't missing any data – we can replace the missing values with observed value to satisfy the null hypothesis equality, i.e., $Y^1 - Y^0 = 0$

Randomized experiment data

Cough frequency for the first six units from honey study under null of no effect

Unit	Potential outcomes		Observed variables		
	Y_i^0	Y_i^1	D_i	X_i	Y_i^{obs}
	cfa			*cfp*	*cfa*
1	(3)	3	1	4	3
2	(5)	5	1	6	5
3	(0)	0	1	4	0
4	4	(4)	0	4	4
5	0	(0)	0	1	0
6	1	(1)	0	5	1

Inference

- Consider some statistic that is a function of the observed variables, D, Y, X, such as the simple difference in means (SDO)

$$\widehat{\delta} = \overline{Y_t} - \overline{Y_c}$$

where $\overline{Y_t} = \frac{1}{N_t}\Sigma_{i:D_i=1} Y_i$ and $\overline{Y_c} = \frac{1}{N_c}\Sigma_{i:D_i=0} Y_i$

- Given a sample of six units, the value of the statistic is

$$\widehat{\delta} = \frac{8}{3} - \frac{5}{3} = 1$$

- Fisher wants to assess how unusual would it be to estimate a 1 under the null hypothesis where there is no effect of the treatment whatsoever.
- The key insight Fisher had was that *we can derive the exact distribution* of $\widehat{\delta}(Y, X, D)$ under the randomization distribution which is the distribution induced by random assignment to the treatment units

Unit	D_1	D_2	D_3	D_4	D_5	D_6	$\widehat{\delta}$
1	0	0	0	1	1	1	-1.00
2	0	0	1	0	1	1	-3.67
3	0	0	1	1	0	1	-1.00
4	0	0	1	1	1	0	-1.67
5	0	1	0	0	1	1	-0.33
6	0	1	0	1	0	1	2.33
7	0	1	0	1	1	0	1.67
8	0	1	1	0	1	0	-0.33
9	0	1	1	0	1	0	-1.00
10	0	1	1	1	0	0	1.67
...							

Conclusion

- If we assign 3 children to the honey, and 3 to nothing, there are

$$\binom{6}{3} = \frac{6 \times 5 \times 4}{3 \times 2} = 20$$

  different assignment vectors (different values for D), and therefore at most 20 unique values for the δ (only ten are given in the table)
- Of these 20 values for δ, 16 were at least as large in absolute value as $\delta(Y, D, X) = 1$, so that the p-value is $\frac{16}{20} = 0.80$.
- At conventional levels (e.g., 0.05), we wouldn't reject the null hypothesis that there is no treatment effect.

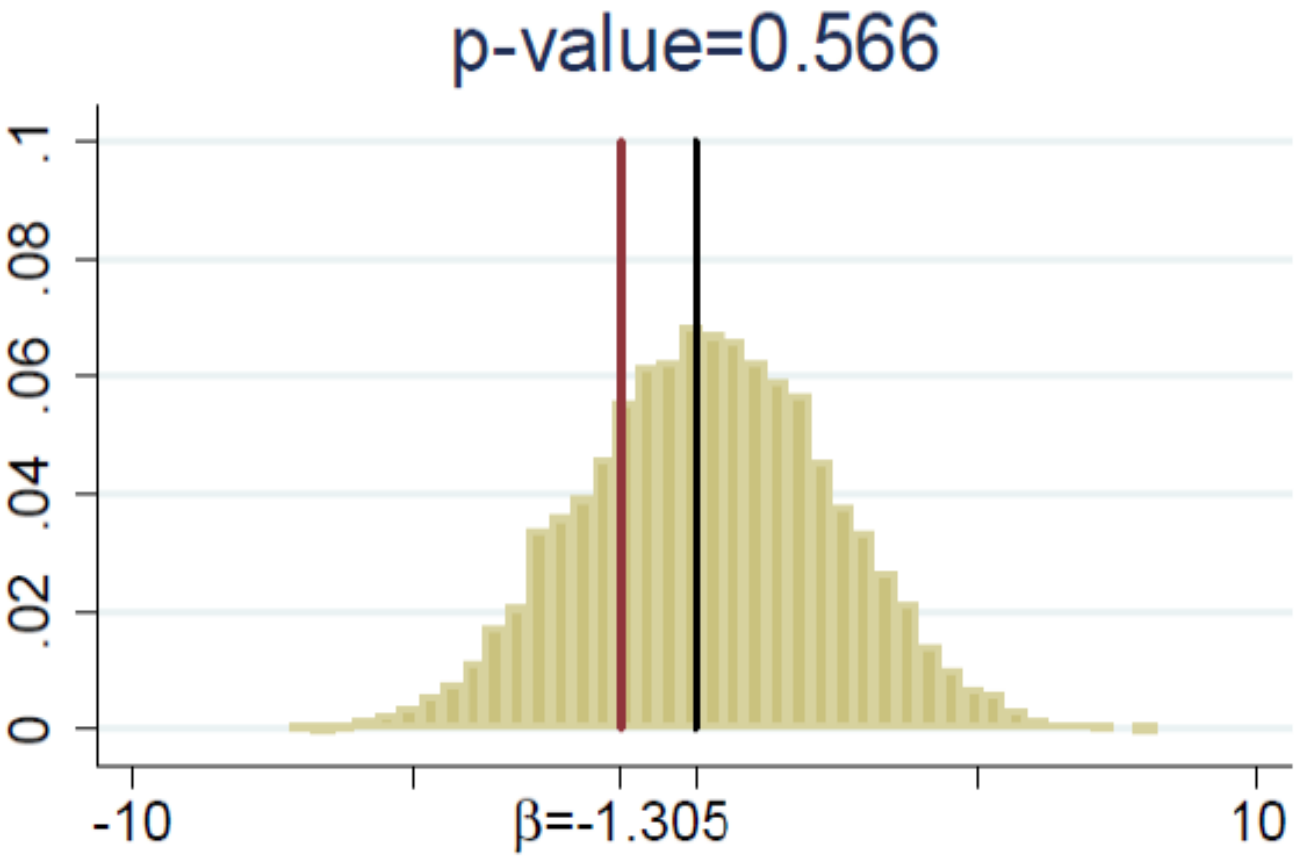
Fisher in today's work

- Useful when sharp null is hypothesis of interest
- Nice feature is that we can produce p values without making assumptions about error variance structure - and without estimating it from our sample
- As a result: preferred method of inference (espacitally in RCTs)

Fisher in practice

- Course of dimensionality
- Consider RCT in 100 schools with 50 getting a treatment
- $$\binom{100}{50} = 1.0089134454556424e + 29$$
- Cannot possible compute exact distribution of outcome under sharp null - too many possibilities
- Solution: choose a random subset of these to approximate sharp null distribution
- Implementation: Take your data and simulate random assignment to geneate outcomes under sharp null.
- Then: compare your experimental estiamte (real world sample) to this distribution

Fisher in practice: Teacher training RCT in schools in England



Example taken from: Murphy, Weinhardt and Wyness (2021) Who teaches the teachers? A RCT of peer-to-peer observation and feedback in 181 schools. *Economics of Education Review*, vol. 82. https://doi.org/10.1016/j.econedurev.2021.102091